

Deep Dive: 21. Modules and Packages

Concept Summary

Modules allow you to split code across multiple files. Any .py file is a module.

Coding Structure

```
import module_name
from module_name import func
```

Example Code

```
import math
from datetime import datetime

print(math.sqrt(16))
print(datetime.now().year)
```

Code Explanation

- Line 1: Imports the entire built-in 'math' module. We must prefix its functions with 'math.'
- Line 2: Imports only the 'datetime' class from the 'datetime' module, allowing us to use it directly without a prefix.
- Line 4: Calculates the square root of 16.
- Line 5: Fetches the current year.

Practice Problems

1. Import the 'random' module and use it to print a random integer between 1 and 10.
2. Create your own file named 'tools.py' with a function, then import it into a different script and run it.